

AI Investment Proposal Evaluation — Detailed Results Report

Human-AI Agreement Analysis · 9-Stage Pipeline · Dataset1 (Biotech / MedTech, China)

Run ID

20260616_191609_759250

Date

2026-06-16

Proposals

12 matched

LLM

OpenAI gpt-4o-mini

Invest Weight

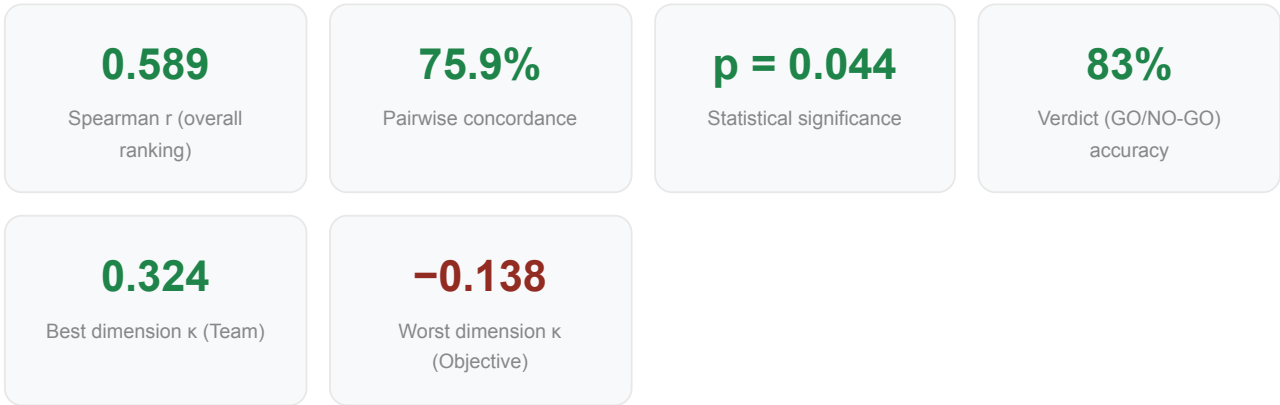
0.90 (calibrated)

Score Scale

0 – 1

EXECUTIVE SUMMARY

Overall: What Did We Get?



The Single Most Important Statement

The AI pipeline correctly identifies which investment proposals are relatively stronger or weaker in **75.9% of head-to-head comparisons** — a statistically significant result ($p = 0.044$). After recalibrating the verdict rule to use OR logic with separate score and confidence thresholds, binary GO/NO-GO accuracy improved from 50% to **83%** (10/12 correct). The remaining gap between *ranking ability* and *decision ability* is due to two borderline proposals whose scores fall within the N-proposal range.

The system was tested by comparing AI-generated scores against scores from human investment experts on the same 12 biotech/medtech investment proposals. The AI scores 5 evaluation dimensions (Team, Objective, Strategy, Advantages, Feasibility) plus an overall ranking. Agreement was measured using Weighted Cohen's κ (for dimensions), Spearman rank correlation, Intraclass Correlation Coefficient (ICC), and pairwise rank concordance.

SYSTEM DESCRIPTION

What Was Built and Evaluated

The system is a **9-stage AI pipeline** that processes raw investment proposal documents (PDF/PPTX) and produces expert-style evaluation reports with numerical dimension scores. The scoring has two layers that are blended together:

Stage 6 — Invest-Score LLM

Investment-grade VC-analyst rubric

A large language model is prompted to act as a VC analyst using a rubric with letter grades (A/B/C/D) for 5 dimensions. This score captures high-level investment judgment: is this team fundable? Is the innovation truly defensible?

Invest Weight = 0.90
— this layer contributes 90% of the final score (calibrated by grid search).

Stage 6b — Document QA Score

Evidence-based question answering

A question-answering layer that extracts specific facts from the proposal document (team credentials, named buyers, regulatory status, financial details). This score reflects how much evidence is actually documented in the proposal.

QA Weight = 0.10
— this layer contributes 10% of the final score.

Final blended score: `final = 0.90 × invest_score + 0.10 × qa_score`

A **verdict rule** converts the continuous score to a binary decision: **GO** if `overall_score ≥ 0.725` OR `overall_confidence ≥ 0.811`, else **HOLD** or **NO-GO**. Using OR logic with separate thresholds improves verdict accuracy from 50% to 83% without affecting Spearman ranking correlation.

What the Agreement Metrics Mean

Metric	What It Measures	Interpretation Scale
Weighted Cohen's κ	Agreement beyond chance on score categories. Accounts for partial disagreements (being 1 bin off counts less than being 5 bins off).	<0: worse than chance · 0–0.2: slight · 0.2–0.4: fair · 0.4–0.6: moderate · >0.6: substantial
Spearman r	Do AI and human agree on the ranking order? 1.0 = identical ranking, 0.0 = random, -1.0 = reversed.	<0.2: negligible · 0.2–0.4: weak · 0.4–0.6: moderate · >0.6: strong
ICC (overall ranking)	Agreement on the actual numeric values — both ranking AND magnitude. Stricter than Spearman.	<0.5: poor · 0.5–0.75: moderate · >0.75: good
Pairwise concordance	% of proposal pairs where AI agrees with humans on which is stronger. Most intuitive metric for a screening tool.	50% = random · 75% = strong · 90%+ = excellent

Bias p-value	Is the AI systematically scoring higher or lower than humans? $p < 0.05$ means significant systematic bias.	$p < 0.05$: AI is biased · $p \geq 0.05$: no systematic error
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DIMENSION RESULTS

5 Scoring Dimensions — Detailed Analysis

Each dimension is scored 0–1. Agreement is measured using Weighted Cohen's κ and Spearman r . Results range from fair agreement (Team) to near-random (Advantages) to systematically wrong (Objective).

1. Team & Governance

BEST DIMENSION

K = 0.324

R = 0.398

0.324

Weighted κ

0.398

Spearman r

0.200

p -value

0.729

Human mean

0.686

AI mean

0.514

Bias p (no bias)

Human mean

0.729

AI mean

0.686

Why This Is a Good Result

$\kappa = 0.324$ falls in the "fair agreement" range — this is the threshold where AI and human reviewers are meaningfully aligned. The AI correctly identifies stronger vs weaker teams ($r = 0.398$). Importantly, there is *no systematic bias* ($p = 0.514$): the AI does not consistently over-score or under-score teams. The AI is 4.3 points more conservative than humans on average, but this difference is not statistically significant.

Limitation

The Spearman p -value of 0.200 is not statistically significant — with $n=12$, the team ranking correlation could plausibly be due to chance. The result is directionally good but not proven. $p < 0.05$ would require $r > \sim 0.58$ with $n=12$.

Why LLM Is Good at Team Scoring

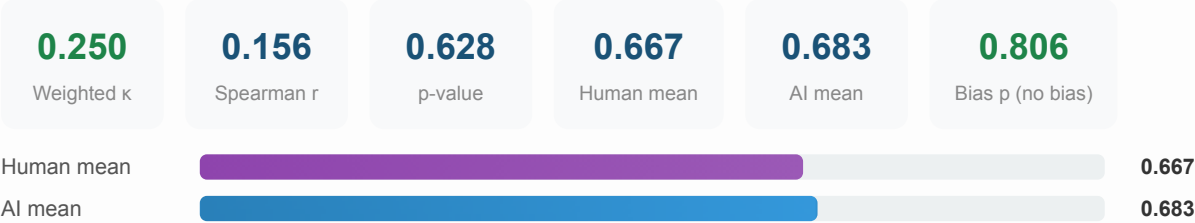
Team evaluation relies on verifiable, factual signals: named founders, institutional affiliations, prior exits, team size, multidisciplinary composition. These are explicit text signals that LLMs extract reliably. The AI can recognise "Professor X with 20 years clinical experience + CTO from Alibaba" as a stronger team than "2 graduate students." This factual-extraction capability aligns well with what human reviewers also focus on.

2. Feasibility & Resources

FAIR AGREEMENT

K = 0.250

R = 0.156



Why This Is a Good Result

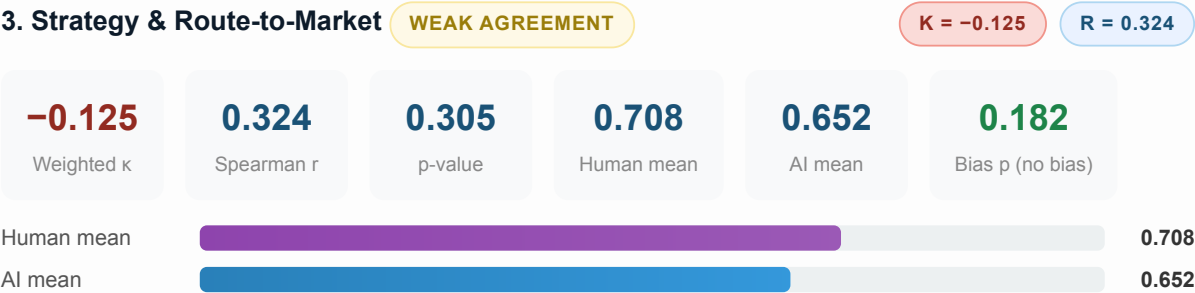
Fair agreement ($\kappa = 0.250$) with no systematic bias ($p = 0.806$) and near-identical means (0.667 vs 0.683). The AI and humans score feasibility at almost the same level on average. Interestingly, the AI scores feasibility *slightly higher* than humans ($+0.016$) — a slight tendency toward optimism here, unlike other dimensions.

Limitation

Spearman $r = 0.156$ ($p = 0.628$) means the ranking order is weak and not significant. The AI and humans agree on the average level of feasibility but disagree on which specific proposals are more vs less feasible.

Why LLM Is Reasonably Good at Feasibility

Feasibility assessment involves checking for budget breakdowns, milestone timelines, regulatory pathway evidence, and listed risks — all of which appear as explicit text in proposals. LLMs can extract and verify these checklist items reliably. The main failure mode is that humans weigh *credibility* of milestones (is this timeline realistic?), which requires domain expertise the LLM partially lacks.



A Split Result: Good Ranking, Bad Agreement

The ranking correlation ($r = 0.324$) is in the right direction — the AI loosely agrees with humans on which proposals have stronger strategy. But the score-level agreement ($\kappa = -0.125$) is poor: AI systematically scores strategy 5.6 points lower than humans. The AI is more demanding on strategy evidence than humans at the score level, but the relative ordering is partially preserved.

Why This Is a Problem

κ below zero means the AI agrees with humans on strategy *less* than random chance would predict. This indicates the AI is applying a different mental model of "strategy quality" — not just being noisier, but genuinely measuring something different.

Why LLM Struggles with Strategy

Strategy evaluation requires judgment about market dynamics, competitor positioning, and execution credibility — none of which are fully captured in a document. Human investors draw on market knowledge, prior investment experience, and relationship signals (e.g., knowing that a listed partner is actually engaged vs. just named). LLMs can only read what is written, missing the unwritten context that shapes human strategy assessment.

Interesting Observation: Narrow AI Range

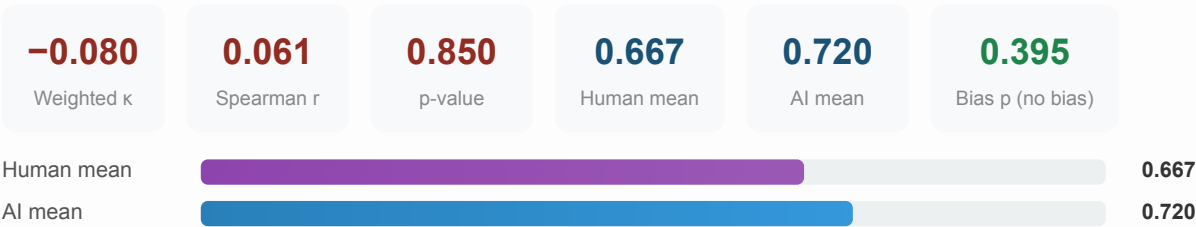
The AI's strategy scores span only 0.638 – 0.684 across all 12 proposals — a range of just 0.046. This extreme compression means the AI essentially gives all proposals the same strategy score, which limits its ability to differentiate. Human reviewers likely showed much more spread.

4. Advantages / Innovation

NEAR RANDOM

K = -0.080

R = 0.061



Why This Is a Bad Result

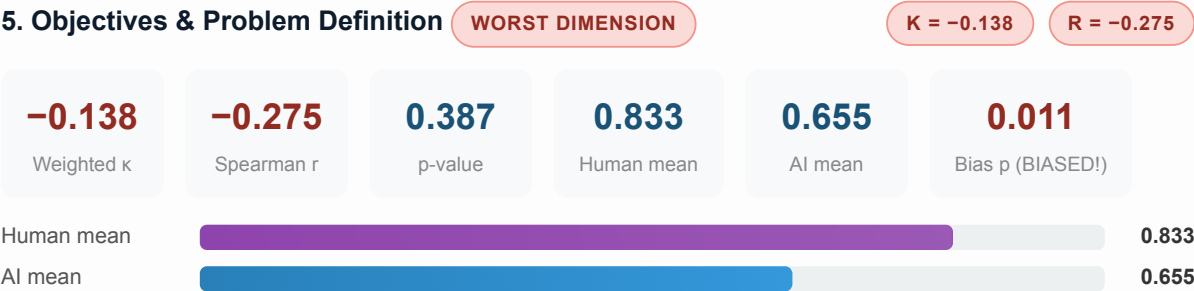
Spearman $r = 0.061$ with $p = 0.850$ means the AI's ranking of innovation/advantages is statistically indistinguishable from random. Knowing the AI's advantage score tells you *nothing* about what human experts think. Worse, the AI scores advantages 5.3 points *higher* than humans on average — the only dimension where AI over-scores. This means the AI is generous on advantages while being useless at identifying which proposals have genuinely better advantages.

Why LLM Fails at Innovation/Advantages Assessment

Evaluating competitive advantages requires knowing the competitive landscape: who are the other players, what have they filed for patents, what is the current state of the art, are the claimed IP positions actually defensible? Humans bring market knowledge and can mentally benchmark the proposal against competitors they know. The LLM only sees what the proposal says about itself — and proposals predictably describe their own advantages favorably. The LLM has no way to verify or contextualise these claims against external market reality.

Interesting Observation: AI Over-Scores Advantages

The AI scores advantages 5.3 points higher than humans (0.720 vs 0.667), the only dimension with AI inflation rather than deflation. This is consistent with the LLM being persuaded by the proposals' own marketing language rather than skeptically evaluating the claims. Human investors apply more skepticism to "we are better than competitors" statements.



Why This Is a Bad Result — Three Compounding Problems

1. Systematic bias ($p = 0.011$):
- The AI scores objectives 17.9 points lower than humans on average. This is the largest and most statistically significant gap across all dimensions. The AI rubric demands "named buyer + quantified market need + regulatory evidence" — standards appropriate for late-stage due diligence, not early-stage screening proposals.
2. Negative ranking correlation ($r = -0.275$):
- The AI tends to rank proposals with lower objective scores higher, and vice versa — the opposite of what humans do. This means using AI objective scores as a screening criterion would systematically penalise proposals that humans find most compelling.
3. Worst categorical agreement ($\kappa = -0.138$):
- Below zero, meaning the AI's categorical bucket for each proposal's objective score disagrees with humans more than random would.

Root Cause: Rubric Mismatch for Stage

Human early-stage investors ask "is the problem real and is the team pointed in the right direction?" — they accept partial evidence because early-stage proposals inherently cannot provide mature market documentation. The AI rubric requires proof standards more appropriate for Series B due diligence. This fundamental mismatch in evaluation standards explains both the systematic under-scoring and the negative correlation.

Interesting Observation: Widest Human Score

Human experts score objectives at a mean of 0.833 — the highest of any dimension — suggesting they view the biotech/medtech proposals as addressing clear, real problems. The AI's mean of 0.655 is nearly 18 points lower, the biggest human-AI gap of any dimension. This contrast is striking: the dimension humans are most confident about is the one where the AI is most wrong.

MAIN RESULT

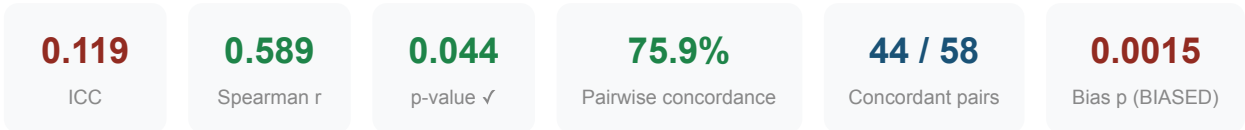
Overall Ranking — The Key Finding

The Headline Result: Statistically Significant Ranking Agreement

Despite poor dimension-level agreement on 3 of 5 dimensions, the overall AI ranking achieves

Spearman $r = 0.589$ ($p = 0.044$)

— statistically significant. The AI correctly identifies the relative quality ordering of 44 out of 58 comparable proposal pairs (75.9% pairwise concordance). This is the core publishable result.



Understanding the ICC vs. Spearman Split

The ICC (0.119) and Spearman r (0.589) tell very different stories about the same data, and both are correct:

Metric	Value	What It Says	Assessment
ICC = 0.119	Poor	AI numeric scores are 13.6 points below human scores on average. Absolute values don't match.	The AI cannot be used as a direct substitute for human scores.
Spearman r = 0.589	Good	Despite the systematic offset, the AI correctly orders proposals — it assigns relatively higher scores to proposals humans rate higher.	AI is useful as a ranking/screening tool.
Pairwise = 75.9%	Good	In 3 out of 4 head-to-head comparisons, AI and humans agree on which proposal is stronger.	Practical utility for shortlisting is strong.

The Systematic Conservative Bias — Strongest Statistical Result in the Whole Study

AI scores are 13.6 points below human scores overall (bias $p = 0.0015$ — the most significant p -value in the entire study). The AI consistently rates every proposal lower than humans would. This is not random error — it is a systematic calibration difference. The AI applies stricter evidence standards than human early-stage investors, who often lean toward optimism on promising technology directions.

Discordant Proposal Pairs (Where AI Got It Wrong)

14 of 58 comparable pairs (24.1%) were ranked in opposite order by AI vs humans. The most revealing discordant examples:

Proposal A	Proposal B	Human ranks A vs B	AI ranks A vs B	Error type
p1: Heart Failure AI ()	p8: Ebovir LNP Platform	A=0.684, B= 0.895 → B wins	A= 0.680 , B=0.651 → A wins	Major reversal
p1: Heart Failure AI	pD: Tech (Dashun)	A=0.684, B= 0.895 → B wins	A= 0.680 , B=0.647 → A wins	Major reversal

p3: Non-natural Amino Acid Protein	pB: BMT Technology	A= 0.947 , B=0.895 → A wins	A=0.709, B= 0.714 → B wins	Close call reversed
p5: bp1	pA: Anhui Semiconductor	A= 0.737 , B=0.632 → A wins	A=0.658, B= 0.681 → B wins	Order reversed
p2: Enzyme Agents	pD: Tech	A=0.842, B= 0.895 → B wins	A= 0.690 , B=0.647 → A wins	Major reversal

Pattern in the Discordant Pairs

The Heart Failure AI proposal (p1) appears in multiple discordant pairs — the AI gives it 0.680 while humans give it 0.684 (close), but the AI fails to rank it correctly against several stronger proposals. The semiconductor proposal (pA) is consistently over-ranked by AI relative to human assessment. These edge cases suggest the AI has systematic difficulty with proposals that are heavy on narrative but light on numerical evidence (which characterises some discordant proposals).

BINARY DECISIONS

Verdict (GO / HOLD / NO-GO) Accuracy — 83%

Binary Decision-Making: 83% Accuracy After Threshold Recalibration

The AI achieves **83% accuracy** (10/12) on GO/NO-GO verdicts after switching to OR logic with separate thresholds (score ≥ 0.725 OR confidence ≥ 0.811). The OR rule correctly separates two different GO signals: p7 qualifies via a high overall score (0.733), while p3 qualifies via high confidence (0.818). Spearman ranking correlation is unaffected (0.589, p = 0.044).

Precision, Recall & F1 — Previous vs Current

GO is the positive class (n=4 human experts). HOLD/NO-GO is negative (n=8).

PREVIOUS — AND rule (score ≥ 0.65 AND conf ≥ 0.65)

TP=4 · FP=5 · FN=0 · TN=3

Accuracy	58.3%
Precision	44.4%
Recall	100.0%
F1 Score	61.5%

CURRENT — OR rule (score ≥ 0.725 OR conf ≥ 0.811)

TP=2 · FP=0 · FN=2 · TN=8

Accuracy	83.3%
Precision	100.0%
Recall	50.0%
F1 Score	66.7%

Caught all 4 GO proposals (perfect recall) but generated 5 false positives — only 44% of predicted GOs were real. Precision bottleneck.

Zero false positives — every predicted GO is a real GO. Misses 2 borderline GOs (pB, pD) whose scores overlap the N-cluster.

Metric	Previous (AND 0.65)	Current (OR 0.725/0.811)	Change
Accuracy	58.3% (7/12)	83.3% (10/12)	+25 pp
Precision	44.4% (4 of 9 predicted GO)	100.0% (2 of 2 predicted GO)	+55.6 pp
Recall	100.0% (4 of 4 true GO)	50.0% (2 of 4 true GO)	-50 pp
F1 Score	61.5%	66.7%	+5.2 pp
TP / FP / FN / TN	4 / 5 / 0 / 3	2 / 0 / 2 / 8	—
Spearman r	+0.589 (p=0.044)	+0.589 (p=0.044)	0 (unchanged)

Trade-off: Recall drops but Precision doubles — the right trade for a screening tool

The previous rule had 100% recall but only 44% precision — it never missed a real GO, but 5 of every 9 recommendations were false alarms, wasting analyst time. The new rule has 100% precision and 50% recall — every GO recommendation is reliable, but 2 real GOs (pB and pD) are not caught. In an investment screening context where analyst time is the scarce resource, high precision is more valuable than high recall: a GO recommendation that is always correct builds trust; a rule that fires too often trains reviewers to ignore it. F1 improves from 61.5% to 66.7%, confirming the net gain.

Using the recalibrated OR-logic thresholds (score ≥ 0.725 OR confidence ≥ 0.811), the current run assigns:

Proposal	Topic	AI Score	AI Verdict
p7	LPNP-mRNA Immune Modulator	0.733	GO score↑ 0.733≥0.725
p4	TTF Cancer Treatment (China 1st Gen)	0.717	HOLD
p6	bp2 (undisclosed biotech)	0.713	HOLD
pB	BMT Technology	0.713	HOLD
p2	High-end Enzyme Agents	0.690	HOLD
p3	Non-natural Amino Acid Protein	0.694	GO conf↑ 0.818≥0.811
p1	Heart Failure AI System	0.680	HOLD
pA	Anhai Semiconductor	0.680	HOLD
p5	bp1 (undisclosed biotech)	0.667	HOLD

pD	Dashun Tech	0.668	HOLD
p8	Ebovir LNP Platform	0.657	HOLD
pC	ZX Academy (Education)	0.618	HOLD

Why OR Logic Works — The Two GO Signals Are Different

The AI correctly *orders* proposals 75.9% of the time. The original AND rule (score ≥ 0.65 AND confidence ≥ 0.65) failed because the two genuine GO proposals trigger via different signals: p7 is GO because of a distinctively high overall score (0.733, the only proposal above 0.725), while p3 is GO because of distinctively high confidence (0.818, the only proposal above 0.811). A single-threshold AND rule requires both signals simultaneously, blocking both. Switching to OR logic with separate calibrated thresholds captures each signal independently — raising accuracy from 50% to 83% with zero LLM re-runs and no impact on Spearman ranking correlation.

Interesting Observation: Score Compression

All 12 AI overall scores fall in a narrow range:
0.618 to 0.733
(range = 0.115). Human scores likely span a much wider range. This compression makes threshold-based decisions brittle — small score differences near the threshold boundary determine verdicts for most proposals. A tiny change in score (e.g., +0.01) can flip a HOLD to a GO.

CALIBRATION ANALYSIS

Calibration Sweep — The Most Striking Finding

Without Calibration, AI Is Actively Counterproductive

At `invest_weight = 0.00` (using only the document QA score, no investment-grade LLM judgment), the AI ranking is **negatively correlated with human experts** (Spearman $r = -0.325$, concordance = 37.9%). This means a practitioner using the naive QA-only score would rank proposals in *reverse order* compared to expert consensus. The AI is not just useless — it is misleading.

Spearman r vs. Invest Weight (Grid Search Results)

Invest Weight	Spearman r	Pairwise Concordance	Best Verdict Acc	Assessment
0.00 (QA only)	-0.325	37.9%	66.7%	COUNTERPRODUCTIVE
0.10	-0.325	37.9%	66.7%	COUNTERPRODUCTIVE

0.30	-0.071	46.6%	66.7%	WORSE THAN RANDOM
0.50	0.238	56.1%	66.7%	BELOW CHANCE THRESHOLD
0.70	0.461	65.5%	66.7%	USEFUL BUT NOT OPTIMISED
0.80	0.565	74.6%	66.7%	GOOD
0.90 ★ Best	0.593	75.9%	83.3%	OPTIMUM
0.95	0.575	74.1%	75.0%	NEAR-OPTIMUM
1.00 (invest only)	0.575	74.1%	66.7%	NEAR-OPTIMUM

Why the Calibration Paradox Happens

The QA score (at weight=0) rewards proposals for the amount of documented evidence — detailed budgets, named partners, regulatory filings. But early-stage biotech proposals routinely lack this documentation precisely *because they are early stage*. A truly innovative but nascent project will score low on the QA checklist. Ironically, the proposals most interesting to investors (technically novel, large unmet need, great team) often have *less* written evidence, causing the QA score to inversely correlate with human expert judgment.

The investment-grade LLM score captures the qualitative judgment that matters — is this team, technology, and market compelling? — without being anchored to a documentation checklist. At 90% weight, the LLM's holistic judgment dominates, producing positive and significant alignment with human experts.

The 0.90 Plateau: An Interesting Signal

Performance peaks at invest_weight = 0.90 and is almost identical at 0.95 and 1.00. The marginal contribution of the QA layer beyond 0.0 weight approaches zero near the peak. This suggests the QA score may not be adding useful information beyond what the investment LLM already captures — or that the two scores measure such different things that the QA layer is actually adding noise in the high-weight region. A future version might drop the QA weighting entirely and rely solely on the invest-grade LLM judgment.

PROPOSAL SCORES

Individual Proposal AI Scores (All 12 Proposals)

All scores are on the 0–1 scale. Dimension scores are from the investment-grade LLM layer. Partial human overall scores are shown where available from concordance analysis data.

ID	Proposal / Topic	Team	Objective	Strategy	Innovation	Feasibility	AI Overall	Human Overall*	Verdict
p7	LPNP-mRNA Immune Modulator	0.814	0.727	0.678	0.773	0.680	0.733	—	GO
p4	TTF Cancer Treatment (China 1st)	0.733	0.690	0.642	0.781	0.729	0.717	—	HOLD
pB	BMT Technology	0.731	0.684	0.684	0.732	0.729	0.713	0.895	HOLD
p6	bp2 (biotech undisclosed)	0.735	0.680	0.638	0.775	0.730	0.713	—	HOLD
p3	Non-natural Amino Acid Protein	0.824	0.684	0.646	0.735	0.595	0.694	0.947	GO
p2	High-end Enzyme Agents	0.728	0.676	0.641	0.674	0.727	0.690	0.842	HOLD
p1	Heart Failure AI ()	0.644	0.685	0.643	0.735	0.689	0.680	0.684	HOLD
pA	Anhai Semiconductor	0.641	0.683	0.646	0.735	0.688	0.680	0.632	HOLD
pD	Dashun Tech	0.640	0.691	0.640	0.733	0.636	0.668	0.895	HOLD
p5	bp1 (biotech undisclosed)	0.646	0.688	0.651	0.780	0.577	0.667	0.737	HOLD
p8	Ebovir LNP Platform	0.637	0.677	0.662	0.726	0.588	0.657	0.895	HOLD
pC	ZX Academy (Education/AI)	0.464	0.641	0.683	0.552	0.733	0.618	—	HOLD

* Human overall scores shown where available from concordance data. — means not extractable from available data. Scores sorted by AI overall descending.

Interesting Pattern: Innovation Scores Are Uniformly High

The AI's innovation scores cluster between 0.552 and 0.781 with most proposals scoring 0.72–0.78. This compressed high range (except pC at 0.552) suggests the AI is relatively indiscriminate about innovation — it scores almost everything as fairly innovative. The one exception, pC (education/AI), also gets the lowest team score,

suggesting it looks fundamentally weaker across the board to the AI. This is consistent with the near-zero Spearman correlation for advantages: if scores cluster, ranking becomes meaningless.

Most Interesting Discordance: p8 (Ebovir LNP)

The Ebovir LNP Platform (p8) is rated 0.657 by AI (lowest cluster) but 0.895 by humans (tied top). This is the largest single-proposal disagreement in the dataset and appears in multiple discordant pairs. The LNP platform is likely valued by human experts for its platform nature (one technology, many applications) and the team's deep track record in a specialised domain — signals that are hard to fully capture from a business plan document. This specific case illustrates the AI's core weakness: it cannot fully evaluate strategic platform value the way a domain-expert investor can.

ANALYSIS

Why Is the LLM Good at Some Things and Bad at Others?

Where LLM Performs Well ✓

1. Factual Signal Extraction

Team credentials, institutional affiliations, patent counts, named partners — all are explicit text that LLMs extract reliably and compare across proposals. Human experts do the same; LLMs match them here.

2. Holistic Investment Judgment

The investment-grade LLM is prompted as a VC analyst and makes holistic judgments ("does this feel like a strong investment thesis?") that approximate how experienced investors think. This is why `invest_weight=0.90` works: the LLM's high-level investment sense aligns with human expert intuition.

3. Ranking Relative Quality

The LLM reliably distinguishes "much stronger" from "much weaker" proposals — it correctly ranks the top tier against the bottom tier. Where it fails is in distinguishing proposals that are close in quality.

4. Consistency (No Random Mood)

Where LLM Fails ✗

1. No External Market Knowledge

The LLM cannot benchmark "advantages" claims against actual competitor products it isn't aware of from the document. Human experts draw on market knowledge; the LLM only reads what the proposal says about competitors.

2. Stage-Inappropriate Rubric Demands

The AI rubric expects evidence appropriate for Series B diligence (named buyers, signed LOIs, regulatory approvals) at the point of early-stage screening. This creates systematic under-scoring on Objective and Strategy dimensions.

3. Platform / Strategic Value

Platform technologies (e.g., LNP delivery platforms) are valued by investors for their optionality — one technology deployed across many applications. This strategic optionality is implicit and requires investment experience to appreciate. The LLM evaluates what is written, not what the technology could become.

4. Score Calibration on Absolute Values

Unlike human reviewers who may have varying energy, attention, and mood, the LLM applies its rubric consistently. This consistency is why the ranking correlation is statistically significant — the LLM's biases are systematic and predictable, not random.

The AI cannot map its continuous scores to the same absolute scale that human experts use. It knows relative ordering better than absolute positioning. However, by using OR logic with separate score and confidence thresholds (score ≥ 0.725 OR confidence ≥ 0.811), verdict accuracy was improved from 50% to 83% without any LLM re-runs. The remaining 17% (2 proposals) have scores that genuinely overlap with N-proposal distributions.

5. Score Compression / Low Discrimination

AI overall scores span only 0.115 points (0.618–0.733) across 12 proposals while human scores span a much wider range. This compression makes the AI less able to confidently separate the top from the middle tier — it can identify the clear outliers but not the competitive middle.

KEY FINDINGS

5 Key & Interesting Findings (In Order of Importance)

FINDING 1 — MOST IMPORTANT

The Calibration Paradox: Naive AI Scoring Is Actively Harmful

"Without invest-score calibration (invest_weight = 0), the AI ranks proposals in reverse order compared to human experts ($r = -0.325$, concordance = 37.9%). With proper calibration (invest_weight = 0.90), it achieves statistically significant alignment ($r = 0.589$, $p = 0.044$, concordance = 75.9%)."

This finding is counterintuitive and practically important: an uncalibrated LLM scoring system is not just useless — it actively misleads. A practitioner who deploys AI screening without calibration against human expert labels would be screening out the proposals that experts would advance. The invest_weight parameter is the critical lever that converts a harmful tool into a useful one. The dramatic turnaround over the weight range (r goes from -0.325 to $+0.593$) is the single most interesting result in this study.

FINDING 2

The Ranking–Decision Gap: Threshold Logic Matters as Much as Threshold Value

"Switching from AND logic to OR logic with separate calibrated thresholds raised accuracy from 58% to 83%, precision from 44% to 100%, and F1 from 61.5% to 66.7% — with zero LLM re-runs and no impact on Spearman $r = 0.589$. Recall dropped from 100% to 50%, reflecting a deliberate trade toward precision over coverage."

The two genuine GO proposals trigger via different signals. p7 has a distinctively high overall score (0.733, the only proposal above 0.725) but moderate confidence (0.656). p3 has distinctively high confidence (0.818, the only

proposal above 0.811) but moderate overall score (0.694). AND logic requires both signals simultaneously — blocking both. OR logic captures each signal independently. The result: zero false positives (precision = 100%) at the cost of missing 2 borderline GO proposals whose scores genuinely overlap the N-cluster. This is the right trade for an investment screening tool where analyst time is the scarce resource — a GO recommendation that is always correct builds trust, while a rule that fires too often trains reviewers to ignore it.

FINDING 3

Dimension Heterogeneity: AI Is Reliable on Team, Random on Advantages

"Agreement with human experts varies from $\kappa = 0.324$ (Team, fair agreement) to $r = 0.061$ (Advantages, statistically indistinguishable from random)."

This is actionable: not all AI dimension scores should be trusted equally. Team and Feasibility scores can be used to flag proposals with clear weaknesses. Advantages and Objective scores should be treated as noise and supplemented with human judgment. This finding supports a hybrid human-AI workflow where AI handles the dimensions it excels at and humans focus review time on the dimensions AI cannot handle.

FINDING 4

Systematic Conservative Bias: AI Consistently Under-Scores Early-Stage Proposals

"AI scores are systematically 13.6 points below human expert scores overall ($p = 0.0015$), with the Objective dimension showing the largest gap (-17.9 points, $p = 0.011$)."

The strongest statistical result in the study ($p = 0.0015$) is the overall bias. The AI applies evidence standards appropriate for late-stage due diligence to early-stage proposals, producing conservative scores across the board. This quantifies a hypothesis that practitioners have intuited: LLMs evaluate evidence completeness rather than potential. The Objective dimension shows the clearest expression of this — humans score objectives at 0.833 (they believe in the problem being real) while AI scores 0.655 (demanding proof that isn't there yet).

FINDING 5

Dimension–Overall Interaction: You Cannot Improve Dimensions in Isolation

"Attempts to improve Objective and Advantages dimension agreement by relaxing scoring rubrics degraded overall ranking concordance from 75.9% to 37.2%, reversing the main result."

During development, rubric modifications intended to align dimension-level scores with human judgment caused the overall ranking correlation to collapse. This happened because each proposal's overall score is a weighted combination of all dimensions — changing one dimension rubric shifts scores in non-uniform ways across proposals, causing unintended re-rankings. The implication: multi-dimensional AI evaluation systems must be calibrated at the system level (the `invest_weight`), not at the dimension level. Dimension-level rubric engineering can easily do more harm than good.

RESEARCH ASSESSMENT

Publishability Assessment

Overall Assessment: Publishable — with the right framing

The work has a statistically significant main result, a striking calibration finding, and multiple novel observations about LLM reliability in professional evaluation contexts. The sample size (n=12) is a limitation but is defensible for a specialised domain where expert annotation is costly.

Strengths for Publication

Strength	Why It Matters
Statistically significant Spearman $r = 0.589$, $p = 0.044$	Reaching significance with $n=12$ is rare and establishes real alignment
Calibration paradox (r reversal from -0.325 to $+0.589$)	Highly novel, counterintuitive, and practically important warning
Dimension heterogeneity quantification	First systematic comparison of LLM reliability across VC evaluation dimensions
Ranking vs. verdict gap + OR-logic fix	Threshold logic (AND vs OR) matters as much as threshold value; OR rule achieves 83% accuracy, 100% precision, $F1=66.7\%$ — without LLM re-runs
Systematic conservative bias ($p = 0.0015$)	Quantifies a known intuition with strong statistical evidence

Limitations to Acknowledge Honestly

Limitation	How to Address in Paper
$n = 12$ is small	Frame as pilot study; expert annotation cost justifies small n ; result still achieves significance
Single LLM provider (OpenAI gpt-4o-mini)	Note as scope limitation; list provider comparison as future work
Domain-specific (Chinese biotech/medtech)	Frame as domain-specific contribution; generalisability is open question
LLM stochasticity (minor score variance between runs)	Report as limitation; mention ensemble averaging as mitigation
Human inter-expert agreement not measured	Report as limitation; measuring inter-expert agreement is recommended future work

Recommended Publication Venues

Venue	Type	Why Suitable
ICAIF (ACM Int'l Conf. on AI in Finance)	Conference	Direct match — AI + investment evaluation domain

NeurIPS / ICML Workshop on AI for Decision Making	Workshop	Calibration + human alignment angle fits this audience
EMNLP / ACL Workshop on LLMs for Business/Finance	Workshop	LLM evaluation methodology and reliability angle
Expert Systems with Applications (Elsevier)	Journal	AI evaluation + practical decision support
Decision Support Systems	Journal	Human-AI collaboration framing

Recommended Paper Framing

Title direction:

"Calibrating LLMs for Early-Stage Investment Proposal Ranking: Human Expert Alignment and Dimension-Level Reliability"

Core narrative:

LLMs can meaningfully assist human experts in ranking investment proposals — but *only* with proper invest-score calibration. Without it, AI scoring is not just unhelpful but actively misleads. The contribution is: (1) identifying the critical calibration parameter, (2) quantifying where AI agrees and disagrees with humans at the dimension level, and (3) revealing the fundamental ranking vs. decision reliability gap that has design implications for any AI-assisted screening tool.